

DASE7506 MP1 Report

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Reported full-test FP32 BPB: 1.48544

1 Introduction

The assignment is to train a language model from random initialisation on the supplied WikiText-2 splits and BPE-2048 tokenizer, then minimise bits per byte (BPB) on the frozen test split. Evaluation uses independent causal windows of 256 tokens, FP32, and must remain reproducible on CPU. The resource cap is at most five times the baseline’s CPU scoring time, 4 GiB peak evaluation RAM, and 64 MiB of uncompressed inference assets. The starter GPT (width 128, four layers, four heads, 1.09M parameters) scores about 2.10 test BPB.

This report documents one submitted predictor and the evidence required by the guide: a reproduced baseline; a comparison at the same number of processed training targets; an ablation of the mechanisms that actually move BPB; and a critical account of quality–cost trade-offs. All architecture and hyperparameter choices were made on validation. The test split was scored only after a method was frozen.

2 Method

2.1 Model

The submitted model is a decoder-only Transformer with context length 256 and vocabulary 2048. Relative to the starter classroom GPT the block is:

- RMSNorm instead of LayerNorm, and no linear biases.
- Rotary position embeddings (RoPE) instead of a learned absolute position table.
- A GELU MLP with hidden width $4 \times d$, matching the baseline FFN family rather than a gated SwiGLU.
- Dropout 0.2 on token embeddings, attention, and residual branches (training only).
- GPT-2-style residual-output initialisation, $\sigma = 0.02/\sqrt{2L}$.
- Tied input and output embeddings (already in the baseline).

The submitted size is $d=256$, $L=8$, 4 heads: 6 820 096 parameters (26.0 MiB checkpoint).

2.2 Training

Training uses the supplied trainer with additional, default-off flags. The submitted run uses CUDA with BF16 autocast and seed 17, which was fixed in advance rather than selected after seeing other seeds. It trains for 7000 steps at batch 128 and context 256, i.e. 229 376 000 processed targets (≈ 63 epochs of the 3.61M-token train split). The optimiser is AdamW with $\beta = (0.9, 0.999)$, gradient clip 1.0, peak learning rate 4×10^{-3} , 300-step warmup, and a cosine floor of 2% of peak. Weight decay is 0.3 and excludes 1-D parameters (norm gains). An EMA copy with decay 0.9995 is validated and saved.

Table 1: Reproduced starter GPT on the official baseline recipe (CPU, four threads, seed 17).

Split	BPB	Token PPL	Parameters	Train tokens
Validation	2.0711	79.5	1 088 256	9 830 400
Test (CPU FP32)	2.10126	80.8	same	same

Table 2: Validation BPB at 9.83M targets (mean $\pm$ sample SD, three seeds). Δ is paired against the modern block.

Config	Params	Val. BPB	Δ vs modern
Classroom GPT	1 088 256	2.0747 ± 0.0037	+0.237
Modern (RMSNorm+RoPE+SwiGLU, no bias, scaled init)	1 049 216	1.8379 ± 0.0021	0
RoPE $\rightarrow$ learned positions	1 081 984	1.9955 ± 0.0047	+0.158
SwiGLU $\rightarrow$ GELU	1 049 728	1.9311 ± 0.0010	+0.093
No residual scaled init	1 049 216	1.8426 ± 0.0022	+0.005
RMSNorm $\rightarrow$ LayerNorm	1 050 368	1.8343 ± 0.0002	-0.004
Add biases	1 054 504	1.8353 ± 0.0024	-0.003

3 Reproduced baseline

The official recipe (starter classroom GPT, 1 200 steps, batch 32, seed 17, AdamW LR 10^{-3} , 100-step warmup, cosine to 10% of peak) was run on CPU with four threads, as in the code README. Table 1 reports the reproduced scores.

This matches the advertised ≈ 2.10 test BPB. Scoring took 7.5 s on the development CPU (Xeon Gold 6248, shared host); peak RSS was 2.11 GiB. GPU replays of the same architecture and token budget (Section 4) give validation BPB 2.0747 ± 0.0037 over three seeds, so the CPU/GPU gap is noise.

4 Matched-token comparison and architecture ablation

The guide asks for a comparison at the same number of processed training targets, and an ablation of the key mechanism. One paired sweep covers both.

Keeping $1\,200 \text{ steps} \times 32 \times 256 = 9.83\text{M}$ targets, seed $\in \{17, 18, 19\}$, and the baseline optimiser, each architectural switch was turned on or off independently. Table 2 and Figure 1 summarise validation BPB.

Under a short pass over a 3.6M-token corpus, replacing the starter block with a modern block cuts validation BPB by 0.24, far larger than seed scatter (SD $\approx 0.002\text{--}0.005$). The two mechanisms that explain almost all of that gap are:

1. RoPE (≈ 0.16 BPB). A learned $256 \times d$ position table must be estimated from few unique windows; late positions are rare. RoPE encodes relative offset in the attention logits with no extra parameters, which is a better inductive bias when data are scarce.
2. SwiGLU (≈ 0.09 BPB). At matched parameter count (hidden width $8/3d$ vs $4d$ GELU), a gated MLP is a more efficient feed-forward nonlinearity—on this short schedule.

Scaled residual init is a small, stable gain (≈ 0.005 , about $3\times$ the paired SD). RMSNorm versus LayerNorm, and removing bias, are not distinguishable from noise at this size; LayerNorm is even slightly better. Those two knobs were therefore treated as optional modern defaults, not as claimed contributions.

CPU scoring of the modern 1.05M model was 7.2 s versus 6.4 s for the baseline on the same host: RoPE adds a rotation, but the ratio is $\approx 1.1\times$, far inside the $5\times$ cap.

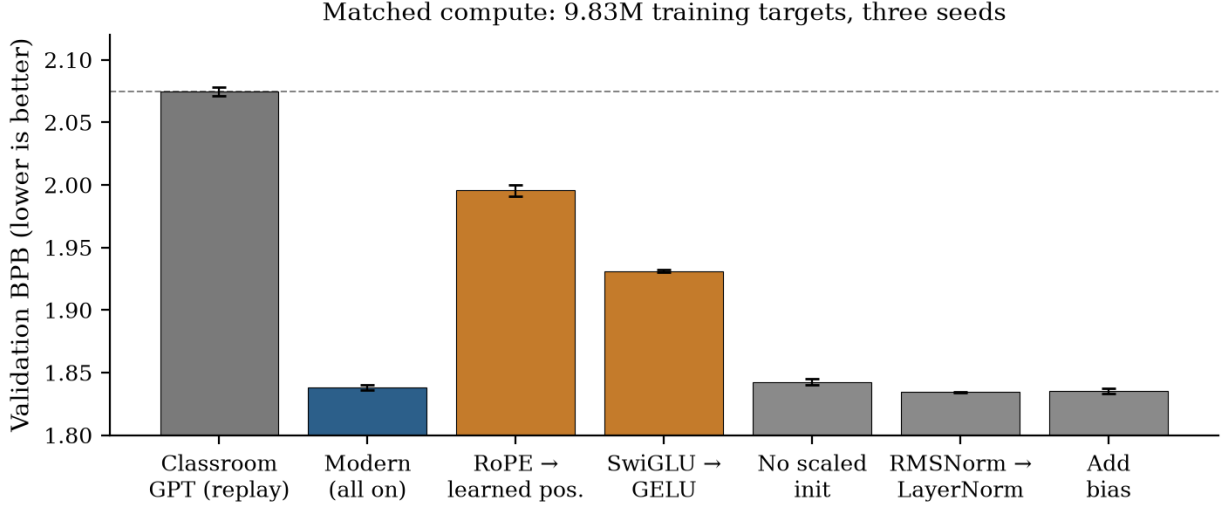


Figure 1: Matched-token architecture ablation. Error bars are sample standard deviations over seeds 17/18/19. The dashed line is the classroom GPT mean.

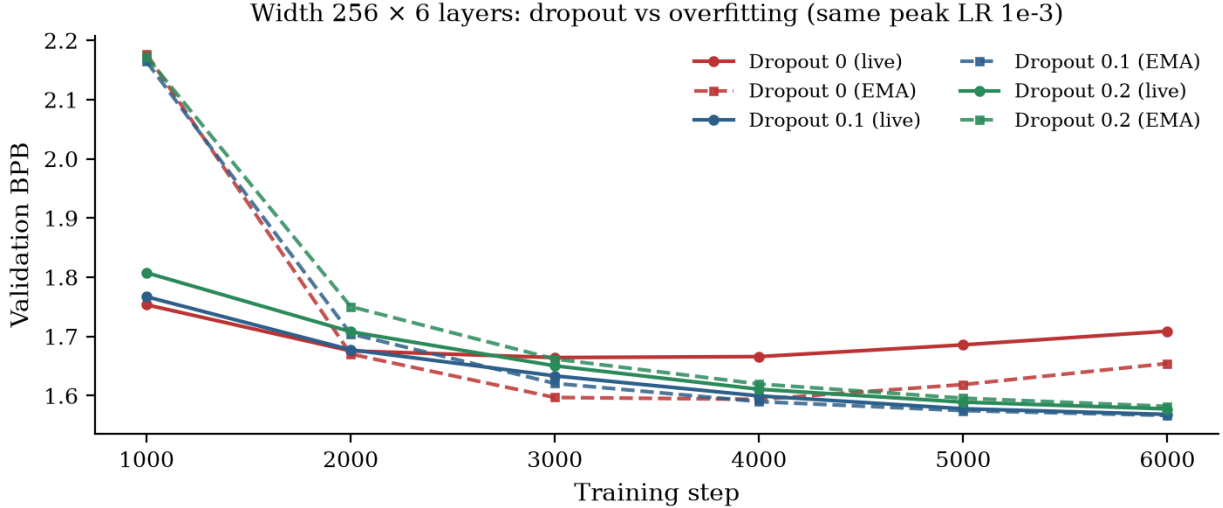


Figure 2: Width 256×6 layers: dropout versus overfitting at peak LR 10^{-3} . Solid: live weights; dashed: EMA.

5 Scaling and regularisation

WikiText-2 is small. The matched-token modern model is still under capacity. The remaining budget (CPU time, 64 MiB, 4 GiB RAM) was used to scale width and depth and to train longer, with regularisation to fight the resulting overfitting. The submitted size is width 256×8 layers: about $3.4\text{--}3.7\times$ baseline CPU scoring time and 26 MiB, inside the cap with a margin for a busy host.

5.1 Dropout is not optional once training is long

Figure 2 holds architecture at width 256×6 layers and varies dropout over 6 000 steps (EMA 0.999). With dropout 0 the live validation BPB bottoms near step 3 000–4 000 (≈ 1.66) and then rises; EMA conceals part of the rise but still degrades after step 4 000. Dropout 0.1 continues to fall to 1.566. Dropout 0.2 is slightly worse at this short horizon (underfitting) but becomes the better regulariser at the final length (Section 6).

Table 3: Final-scale paired ablations (229M targets, seed 17).

Variant	Val. BPB	vs SwiGLU run
SwiGLU, wd 0.1 (previous freeze)	1.4870	0
Same run, no EMA (live weights)	1.5036	+0.017
RoPE $\rightarrow$ learned positions	1.5311	+0.044
Entire block $\rightarrow$ scaled classroom GPT	1.5153	+0.028
Dropout 0	2.214 (live 2.985)	+0.73
SwiGLU $\rightarrow$ GELU, wd 0.1	1.4779	-0.009
GELU, wd 0.3 (submitted)	1.4672	-0.020

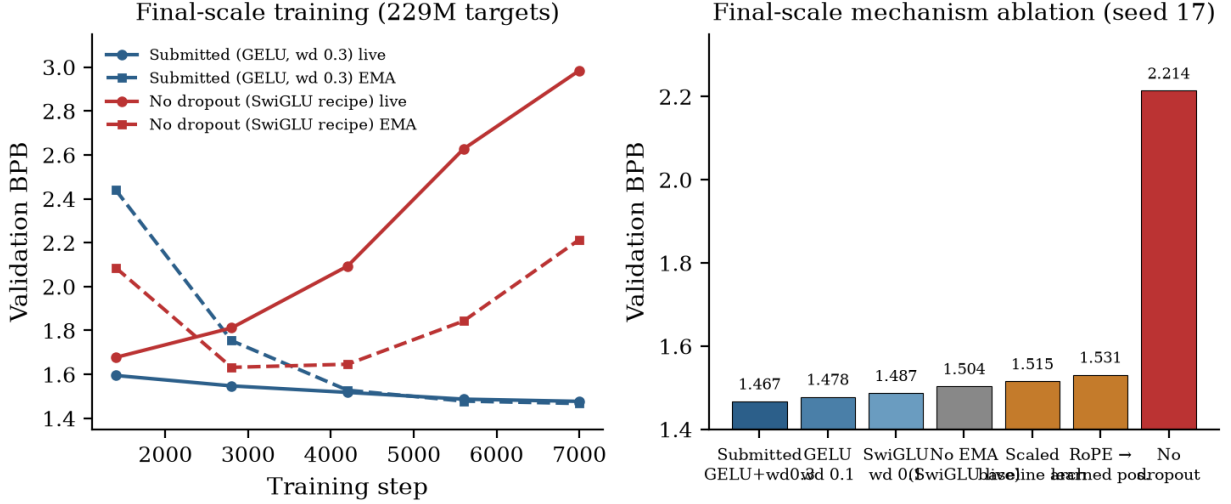


Figure 3: Left: live versus EMA validation BPB for the submitted recipe and a no-dropout control. Right: final-scale mechanism ablations (seed 17).

6 Final-scale ablation and the SwiGLU reversal

At the submitted token budget (229M targets, batch 128, LR 4×10^{-3} , EMA 0.9995, dropout 0.2, seed 17), mechanisms were removed one at a time (Figure 3, right). Table 3 reports EMA validation BPB unless noted.

Dropout is the dominant regulariser: without it the live model diverges after ≈ 1.4 k steps (Figure 3, left). EMA is a cheap averaging of the optimisation path and is worth ≈ 0.01 – 0.017 BPB at this scale; it is a training-time copy, not a test-time ensemble, so scoring cost is unchanged. RoPE remains helpful but the gain shrinks from 0.16 (short training) to 0.04 (long training): more epochs let a learned table catch up, yet not completely. Replacing the whole modern block by a scaled classroom GPT, trained with the same recipe, lands at 1.515: most of the journey from 2.07 to 1.47 is scale + schedule + dropout + EMA.

SwiGLU reverses sign. At 9.83M tokens SwiGLU won by 0.09; at 229M tokens with dropout 0.2, GELU wins by ≈ 0.014 on average over three seeds (GELU 1.4765 ± 0.0026 vs SwiGLU 1.4904 ± 0.0036). A gated MLP is a higher-capacity FFN; once the model is large and the data are seen ≈ 60 times, that extra capacity overfits even with dropout.

GELU + weight decay 0.3 is the submitted method. It does not change scoring time relative to a SwiGLU 256×8 checkpoint. CPU scoring of the submitted predictor is about 3.4 – $4.1 \times$ the local baseline (quieter host $\approx 3.4 \times$), peak RSS is 2.10 GiB, and the uncompressed checkpoint is 26.0 MiB, all inside the evaluation caps.

7 Training and search cost

Development used one RTX 3080 Ti (12 GiB). About 68 recorded GPU training runs sum to 29 800 s $\approx$ 8.3 GPU-hours of job time (many overlapped on one GPU, so wall-clock was shorter). Additional CPU time: the official 1 200-step baseline ($\approx$ 7 min), repeated scorer timings, and all ranked FP32 evaluations. No external text, no pretrained weights, no paid API.

8 Critical analysis

What worked. (i) RoPE is the one architectural change that helped at both token budgets. (ii) Dropout plus a long cosine schedule is what makes a 6.8M model viable on 3.6M tokens. (iii) EMA is a one-line, zero-inference-cost improvement that should be in any ablation table because the trainer already holds the live weights. (iv) GELU + stronger weight decay beat a fashionable gated MLP once training was long—an empirical correction, not a slogan.

What did not. RMSNorm and dropping bias did not matter at 1M parameters and were not re-litigated as “wins.” Extra heads, extra width at fixed time, and simply training longer without more decay all failed. A 9-layer net used more of the $5\times$ budget for no validation gain: the bottleneck is data, not FLOPs.

Trade-off. Moving from the 1.05M modern GPT (val. 1.84) to the 6.8M submitted model (val. 1.47) costs about $3\times$ scoring time and $6\times$ parameters, still inside the cap. Further scale would spend the remaining $1\times$ time for a likely sub-0.01 BPB and a higher overfitting risk.

Limitations. Several late ablations used a single seed; the GELU and weight-decay conclusions were replicated on seeds 18 and 19. The CPU time ratio is host-load dependent; the architecture ratio measured on a quieter interval is $\approx 3.4\times$. Substantive implementation and drafting help from a coding agent (Cursor Grok 4.6) is disclosed in the repository README.

9 Conclusion

The submitted predictor is a 256-wide, 8-layer causal GPT with RMSNorm, RoPE, GELU, dropout 0.2, and EMA weights, trained for 229M targets with AdamW weight decay 0.3. It scores 1.48544 full-test FP32 BPB against a reproduced baseline of 2.10126, within the three evaluation limits. The matched-token study isolates RoPE and (under short training) SwiGLU; the final-scale study shows that regularisation and training length dominate architecture, and that SwiGLU hurts once the model is large. Those two controls together satisfy the guide’s evidence requirement.

References

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